

QUESTION BANK CAE-1

Subject: [Operating Systems]

Class: SYAI[A] (UCAIL202)

Blooms Taxonomy level

1: Remember, 2: Understand, 3: Apply, 4: Analyse, 5: Evaluate, 6: Create

Q. No.	Question	Marks	BL	CO	PO
UNIT I					
1	Explain time sharing operating system.	4	2	1	1
2	Explain Real time OS with the help of a diagram. List its type.	4	2	1	1
3	Describe the evolution of the operating system.	4	2	1	1
4	What is a multiprocessor system ? Give two advantages to it.	3	1	1	1
5	What is a system call ? Enlist any four system calls.	3	1	1	1
6	Describe six services provided by an operating system.	3	2	1	1
7	Explain layered and microkernel operating system structure.	4	2	1	2
8	Describe the purpose of system calls with Explain how parameter passing is done while implementing system calls.	4	2	1	2
9	Discuss the Basic hardware support necessary for modern operating systems.	4	2	1	5
10	Describe following operating system structures. i) Monolithic ii) Microkernel.	4	2	1	1

UNIT II																				
11	What is the process ? Explain the different process state with the help of a state diagram.	4	1	2	1															
12	What is multithreading ? Explain with a suitable diagram.	3	1	2	2															
13	State and describe types of schedules. Describe how each of them schedule the job.	4	2	3	2															
14	Differentiate between short term and long term scheduling.	4	3	3	1															
15	What is pre-emptive and non preemptive scheduling?	4	1	3	2															
16	Calculate average waiting time for following data using First Come First Served (FCFS) and Shortest Job First (SJF) algorithms <table><tr><td>Process</td><td>Burst Time</td><td>Arrival Time</td></tr><tr><td>P0</td><td>8</td><td>0</td></tr><tr><td>P1</td><td>4</td><td>1</td></tr><tr><td>P2</td><td>5</td><td>2</td></tr><tr><td>P3</td><td>3</td><td>3</td></tr></table>	Process	Burst Time	Arrival Time	P0	8	0	P1	4	1	P2	5	2	P3	3	3	4	3	3	2
Process	Burst Time	Arrival Time																		
P0	8	0																		
P1	4	1																		
P2	5	2																		
P3	3	3																		
17	Describe one-to-one multithreading model with suitable diagram. Also write any two advantages of one-to-one model over many-to-one model.	4	2	2	2															
18	What is Scheduling? State and explain Three scheduling criteria.	4	1	3	2															
19	Describe I/O Burst and CPU burst cycle. Give one example.	4	2	3	2															
20	Differentiate the process and thread on four points.	4	3	2	2															

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